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Abstract - We have often heard of cases of loss of animals in the world, the owners of these animals are very troubled in looking for lost animals, so often when the pet is lost, the owner leaves it or makes the news of the competition for who can find it. We strive to help these pet owners easily detect the presence of pets they have through a mobile application that we make, namely Detepet that can track all the pets you have, so you are not afraid of losing your pet. In this case, the user should register their pet and the tool GPS necklace will be attached to the pet and the pet will be under monitoring of the systems.

Keywords: Tracking mobile pet application, detecting animals application, pet detect mobile application

I. INTRODUCTION

People who have pets must love these pets so much that they must take care of them. However, when the pet is lost, the person who has the pet must be troubled when he wants to find the pet. The impact of losing these pets is that the pet owner feels sad and worried about the pet he has cared for. The high number of cases of loss of pets is caused by recalcitrant pets and negligence of the pet owner. These cases can be overcome in various ways, such as increasing awareness of pets.

For the case of losing these pets, according to the results of the study, it was found that people who experience the loss of pets that are kept for years deeply feel sadness in their lives and can hurt them. This is because pets that have been kept for years have a very close relationship with their owners, become ideal friends, and teach many things to the owner. So prevention for cases like this should be done immediately so that people who have pets when the pet is gone, don't panic and feel deep sadness. Jakarta Animal Aid Network says that there are 100 million pets lost within a year in Indonesia[1].

In this day and age, there are a lot of people who keep pets to play with or keep a house when all family members leave home. We have heard of losing pets too often, especially pets such as dogs and cats. The most severe case was the dog loss because he wanted to find his son who was hit by a car on the highway.

Because of the large number of cases of missing pets, people who love pets make the community care about lost pet stars or are tortured by their owners. This community is called Jakarta Aid Animal Network. In this community, we can share about pets, such as news of losing pets, open adoption for pets, sales of pet goods, and so on. People in this community also respond quickly to lost news about pets.

Understanding people who see lost pets is needed so that the owner does not have trouble finding a lost pet. Social media can be a platform for communication between the owner and the inventor of the pet so that cases of loss of pets can be resolved immediately. In this day and age, the role of smartphones is very vital in our lives, but there are still many people who do not understand this sophisticated technology.

People in the future must focus on using technology so that it can facilitate all the activities we do daily. Pet owners are very difficult to control their pets, starting from how many steps do pets live every day, how many pet heart beats, how much oxygen they breathe per minute, and much more. With the existing technology, it is expected to be able to help all problems from pet owners[2].

Pet owners find it difficult to track the health of their pets and look for their pets when they are lost. With the Detepet mobile application, we hope that pet owners can overcome all the problems described above, from losing pets, measuring heart rate, measuring breathing rates to pets, and counting footsteps that pets live every day.

Detepet is also expected to be a solution to the problem of all types of pets. Pet owners who use Detepet are also expected to be comfortable using this application so that the community for the Detepet application is expanding and covering all of Indonesia.

II. PREVIOUS RESEARCH

There are several previous research applications such as Detepet, such as the Paw Track, Tractive, Pod, TabCat. The application also helps to track or track pets by wearing a

necklace on the neck of a pet who wants to track it. Besides, there is also a tool called Rimas Cod which is a necklace that is worn on our pets, and we can squeeze a remote, then the necklace worn on our pets goes off, so we know its position. Applications to track or track pets use technology, such as GPS (Global Positioning System), internal Wi-Fi, and RFID (Radio Frequency Identification). GPS is a very good technology for tracking these pets. With GPS, we know the exact position of our pets, by showing the coordinates when using the application. However, none of the applications and tools above can track the health of pets, and the footsteps of these pets. Therefore, we want to make an innovative application and tool that is very useful for existing pets[3].

There is some previous technology which applies to help human to give well being for their pets including having pets as a virtual pet.

- 1 Using Arduino system for pet's food feeder [4] which is in a food storage container and including pet's water dispenser which can measure how much water for a pet on daily basis[5]. Moreover, apply the pet litter box which can measure how much pet defecation [6][7].
- 2 A mobile application was developed in the zoo using 3 different languages such as English, Arabic, and French where the zoo's visitor can attach the animal cage's QR tag and get information regarding the animal either text or voice as text to speech. Moreover, the application can play an animal sound based on the attached QR tag and including the zoo's visitor can buy a pet in the zoo and rating the application[8].
- 3 A sensor GPS was attached to the pet and used to track the pet's location around the house.[9]. Another pet's tracking system application, named Bokk Meow was implemented in Thailand and include six main application functions [10]
- 4 Another research applies to poultry pet such as chicken, duck, and so on, and using special pet jacket which attached with a microcontroller in order to detect physical touch media as human-animal interaction systems[11].
- 5 in the Philippines, a mobile application for monitoring pets was proposed using sensor and designed remote activated smart pet door for pets' house such as cat and dog including defecation pad, food, and water dispenser [12]. Moreover, the presence of the pet's owner can be represented as voice activation and the owner can set the room temperature, voice activation, and camera services through a webcam.
- 6 Meanwhile, a mobile application was proposed as well for pets monitoring health such as dog and cat which was created for pet's disease prevention and this application is connected automatically to veterinarian including to find the nearest pet's hospital and clinic [13].
- 7 Dog as a pet was tracked to check the dog's health condition regarding their daily activity including

ransom food and water to show the pet's owner regarding with dog's welfare [14].

- 8 Another mobile application called Virtual Internet Pets (VIP) was proposed for those who are interested to have a virtual pet that can be raised online and developed using java based mobile agent as a computer interface to simulate life-like pets as well[15].
- 9 The Augmented reality application based on hand/palm pose recognition was built in order to present the favorite's virtual pet for each person on their palm and can play and interact with their pet using their hand and finger naturally[16].
- 10 The animal tracking system using Arduino was proposed using a simple analog light sensor with motion logic and various sensors with GPS and public Google Maps which can show the right location and position of the appointed animals[17].
- 11 Meanwhile, a tracking system was proposed to track not only pets but for disabled, children, and elderly people where this system was built using GPS, GSM network, and GPS satellite which can transmit the longitude and latitude coordinates from the appointed object including pets[18].
- 12 Pet monitoring was applied using animal biometric identifier which uses face's pet as biometric identification and in this case, just limits to pet such as dog and this is applied as monitoring pet animal in the smart city which can detect every each roaming pet in the city [19].

III.PROPOSED IDEA

We make pet tracking applications and have many useful features for pet owners who use the Detepet application, especially for people who like to lose pets, so they are not sad and panic anymore when losing pets happens, just by using this application all problems regarding animals can occur.

For the user interface, we use Axure RP as the initial display of the Mobile Detepet Application, and also we use Android Studio to make applications on the Android and XCode platforms for applications on the IOS platform.

The following are the features that we provide in the Detepet Mobile Application:

- Promotional Videos

In the application that we made, at the beginning of the application, you will be shown facts about your pet, as well as advertisements about pets.

- Pet Tracking

The application that we made has the main goal of tracking pets so that cases of loss of pets can be minimized. Tracking This pet uses GPS technology, like couriers when sending pictures, on the application we can see where our goods are. This tracking also has the same system as the courier. This Pet Tracking must be supported by a device in the form of a

necklace worn on the neck of a pet, so that the device can be connected to the Detepet mobile application, and can track pets wherever they are.

- Forum

The application that we made has a forum feature to discuss with each other and exchange ideas with people who share similar hobbies. This forum will be categorized according to pets maintained, such as cats, dogs, hamsters, birds, and many more. This forum was created so that those who share the same hobby can become more familiar and eventually meet each other with each pet.

- Customer Service 1x24 hours

This application is supported by customer service that serves 24 hours so that if there are questions and complaints in using the Detepet application they can be resolved immediately.

- Pedometer



Fig. 1. User Interface Detepet Mobile Application

The application that we made is supported by pedometer technology. A pedometer is a device, in modern times it can usually be portable and designed electronically or electromechanically, which calculates every step of a pet by detecting the body movements of the pet. Because the distance between each pet varies, informal user calibration is needed if standardization of step distances (such as kilometers or miles) is desired. So that we can see how many steps our pets have walked every day.

- Cardiotocography (CTG)

The application that we made is also supported by Cardiotocography technology. CTG is a tool used to monitor heart rate. CTG technology is also used on smartphones today.

- Information

Our application provides a variety of information about the news of the loss of pets, how to care for pets according to the pets we choose.

- Event

Our application provides information about events around pets along with the schedule and place of execution. The usual events are pet fashion, birds chirping competition, and many more.

- Open Adoption of Pets

Our application provides open adoption for fellow hobbyists. The requirement to become an adopter will be explained when we open the Detepet application.

- Online store

This application is also supported by an online store for the sale of various tools regarding pets, and pet food itself.

The Detepet Application Mobile Use Case diagram as seen in figure 1 contains various things that can be done for people who use this application. People who want to use the Detepet Application must register first before using the application including register their pet as well. After that they can make the necklace item order form provided by the Detepet application, then after making the order form, we must make a payment form to continue the purchase of the Detepet necklace transaction. After all, is paid, the system will verify and estimate the items sent within 3 working days. After buying, the customer can create a pet profile that wants to be tracked, after creating a profile, the customer chooses a subscription. This subscription determines the range of pets that can be tracked. Customers can also post information in this application. Each order can have feedback and rating where the user can write their order experiences and this is very powerful for the next transaction from other customers to recognize our services as excellent services. Moreover, Forum is one of tool which was created in order to build communication between user, systems and other users as well and discuss any information and issues regarding with pet and particularly for the systems as well.

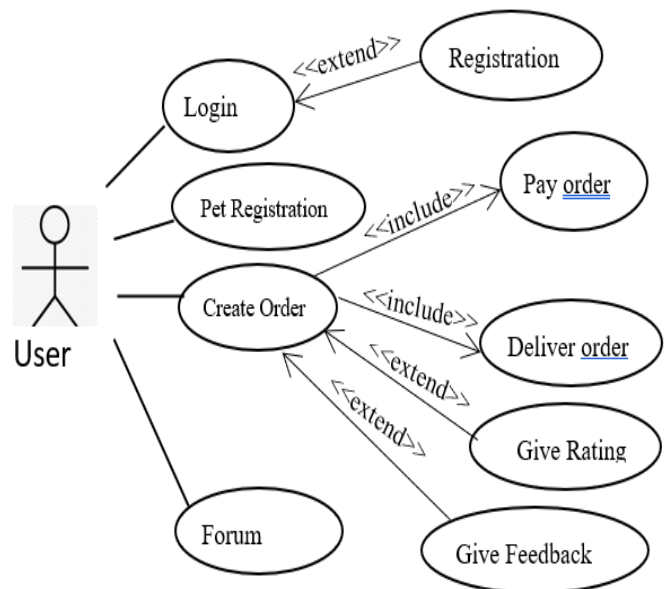


Fig. 2. Use Case Diagram of the proposed model

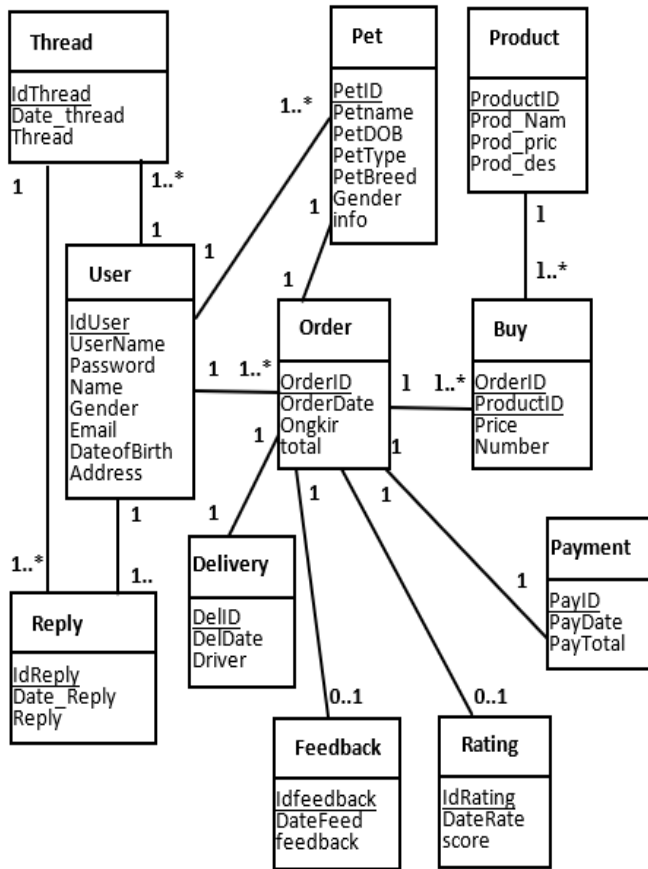


Fig. 3. Class Diagram of the proposed model

Based on Deaf Helper Mobile Application Use Case in Figure-2. We created a Class Diagram, which contains 11 classes in total, including user, order, thread, reply, pet, product, buy, delivery, payment, feedback, and rating. This class diagram is a data class diagram which a representation of database implementation which uses 11 tables database.

As seen in figure 1 where the user should log in and if they are a new user then they should do the registration and their data will be recorded in class user. Moreover, the user should register their pet and their pet's data will be saved in class pet and each user can register as many as their pet. When the user wants to have a necklace for tracking their pet then they should order and decide which kind of necklace tracking that they want. This system is also possible for the user to buy any product available in the store online retail. This order transaction will include class user, order, buy, product and pet.

After the order then the user should do the payment with any available payment such as debit card or credit card and possibly for COD (Cash on Delivery) as well to make easy the payment as user's interest. After the payment, then the store will assign for delivery and the bill for delivery will be recorded as well and include the total payment. If the user comes to the store then the delivery bill will be zero. The payment and delivery transactions as seen in the use case diagram in figure 1 as <<include>> activity which means should be done. Finally, the user can give their feedback and

rating which will be saved in classes feedback and rating respectively. The feedback and rating transactions as seen in the use case diagram in figure 1 as <<extend>> activity which means optional where the user can give or not for their feedback and rating.

The forum activity as seen in use case diagram figure 1 will use 3 classes such as user, thread, and reply. The thread class will record any raise question or issue which will be questioned and any reply comment will be saved in the class reply. This forum is a very powerful tool for this application since this tool, can help to understand any raised problems regarding the transaction and pet issues.

IV.CONCLUSIONS

People who have pets certainly want to always look after pets that are loved and cared for years so there are no cases of loss of pets. Our application is here to answer all problems regarding the loss of pets, we believe this application can help people who often experience the loss of pets. This application can find out where your pet is when you wear the necklace you bought on this application, and connect with this application.

We recommend that people who often experience problems with losing pets can use our application because besides being able to track your pet, this application is also supported by a heart rate and breathing rate detector, as well as a footprint counter for your pet, so you will feel safe and comfortable when away from your beloved pet.

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